In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

In [2]:
```
df=pd.read_csv("iris.csv")
```

In [3]:
```
df
```

Out[3]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	virginica
146	6.3	2.5	5.0	1.9	virginica
147	6.5	3.0	5.2	2.0	virginica
148	6.2	3.4	5.4	2.3	virginica
149	5.9	3.0	5.1	1.8	virginica

150 rows × 5 columns

In [4]:
```
# getting overall information of data( how many colums,columns names, is null va
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        --------------  -----  
 0   sepal_length  150 non-null    float64
 1   sepal_width   150 non-null    float64
 2   petal_length  150 non-null    float64
 3   petal_width   150 non-null    float64
 4   species       150 non-null    object 
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

In [5]:
```
# displaying first five records
df.head()
```

Out[5]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

In [6]:

```
#displaying first 15 records
df.head(15)
```

Out[6]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa
5	5.4	3.9	1.7	0.4	setosa
6	4.6	3.4	1.4	0.3	setosa
7	5.0	3.4	1.5	0.2	setosa
8	4.4	2.9	1.4	0.2	setosa
9	4.9	3.1	1.5	0.1	setosa
10	5.4	3.7	1.5	0.2	setosa
11	4.8	3.4	1.6	0.2	setosa
12	4.8	3.0	1.4	0.1	setosa
13	4.3	3.0	1.1	0.1	setosa
14	5.8	4.0	1.2	0.2	setosa

In [7]:

```
# displaying last five records
df.tail()
```

Out[7]:

	sepal_length	sepal_width	petal_length	petal_width	species
145	6.7	3.0	5.2	2.3	virginica
146	6.3	2.5	5.0	1.9	virginica
147	6.5	3.0	5.2	2.0	virginica
148	6.2	3.4	5.4	2.3	virginica
149	5.9	3.0	5.1	1.8	virginica

In [8]:
```
# dimensions of data
df.shape
```

Out[8]:
```
(150, 5)
```

In [9]:
```
# checking size
df.size # returns result as (rows*columns)
```

Out[9]:
```
750
```

In [10]:
```
# checking dimensions
df.ndim # returns dimension of data
```

Out[10]:
```
2
```

In [11]:
```
# kowning statistics of data
df.describe()
```

Out[11]:

	sepal_length	sepal_width	petal_length	petal_width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.054000	3.758667	1.198667
std	0.828066	0.433594	1.764420	0.763161
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

In [12]:
```
# checking null values
df.isnull().sum()
```

Out[12]:
```
sepal_length    0
sepal_width     0
petal_length    0
petal_width     0
species         0
dtype: int64
```

In [13]:
```
#The index (row labels) of the Dataset.
df.index
```

Out[13]:
```
RangeIndex(start=0, stop=150, step=1)
```

In [14]:
```
#The column labels of the Dataset.
df.columns
```

Out[14]:
```
Index(['sepal_length', 'sepal_width', 'petal_length', 'petal_width',
       'species'],
      dtype='object')
```

In [15]:
```
# checking datatypes in dataset
df.dtypes
```

Out[15]:
```
sepal_length    float64
sepal_width     float64
petal_length    float64
petal_width     float64
species          object
dtype: object
```

In [16]:
```
#Return the columns values in the Dataset in array format
df.columns.values
```

Out[16]:
```
array(['sepal_length', 'sepal_width', 'petal_length', 'petal_width',
       'species'], dtype=object)
```

In [17]:
```
#Generate descriptive statistics to view some basic statistical details like per
df.describe(include="all")
```

Out[17]:

	sepal_length	sepal_width	petal_length	petal_width	species
count	150.000000	150.000000	150.000000	150.000000	150
unique	NaN	NaN	NaN	NaN	3
top	NaN	NaN	NaN	NaN	setosa
freq	NaN	NaN	NaN	NaN	50
mean	5.843333	3.054000	3.758667	1.198667	NaN
std	0.828066	0.433594	1.764420	0.763161	NaN
min	4.300000	2.000000	1.000000	0.100000	NaN
25%	5.100000	2.800000	1.600000	0.300000	NaN
50%	5.800000	3.000000	4.350000	1.300000	NaN
75%	6.400000	3.300000	5.100000	1.800000	NaN
max	7.900000	4.400000	6.900000	2.500000	NaN

In [18]:
```
# Read the Data Column wise
df['species']
```

Out[18]:
```
0         setosa
1         setosa
2         setosa
3         setosa
4         setosa
         ...
145    virginica
146    virginica
147    virginica
148    virginica
149    virginica
Name: species, Length: 150, dtype: object
```

In [19]:
```
#Sort object by labels (along an axis).
df.sort_index(axis=1,ascending=False) #prints data in decending order
```

Out[19]:

	species	sepal_width	sepal_length	petal_width	petal_length
0	setosa	3.5	5.1	0.2	1.4
1	setosa	3.0	4.9	0.2	1.4
2	setosa	3.2	4.7	0.2	1.3
3	setosa	3.1	4.6	0.2	1.5
4	setosa	3.6	5.0	0.2	1.4
...	...	...	...	...	...
145	virginica	3.0	6.7	2.3	5.2
146	virginica	2.5	6.3	1.9	5.0
147	virginica	3.0	6.5	2.0	5.2
148	virginica	3.4	6.2	2.3	5.4
149	virginica	3.0	5.9	1.8	5.1

150 rows × 5 columns

In [20]:

```
#sort values by column name
df.sort_values(by="species")
```

Out[20]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
27	5.2	3.5	1.5	0.2	setosa
28	5.2	3.4	1.4	0.2	setosa
29	4.7	3.2	1.6	0.2	setosa
30	4.8	3.1	1.6	0.2	setosa
...	...	...	...	...	...
119	6.0	2.2	5.0	1.5	virginica
120	6.9	3.2	5.7	2.3	virginica
121	5.6	2.8	4.9	2.0	virginica
111	6.4	2.7	5.3	1.9	virginica
149	5.9	3.0	5.1	1.8	virginica

150 rows × 5 columns

In [21]:

```
#Purely integer-location based indexing for selection by position.
df.iloc[5] # gives data values that are present at 5 th index
```

Out[21]:
```
sepal_length       5.4
sepal_width        3.9
petal_length       1.7
petal_width        0.4
species         setosa
Name: 5, dtype: object
```

In [22]:
```
#selecting via [], which slices the rows
df[0:3] # excludes last number-> [0:3] will print rows from 0 to 2 index
```

Out[22]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa

In [23]:
```
#selection by label( selecting only particular column)
df.loc[:,["species","petal_length"]]
```

Out[23]:

	species	petal_length
0	setosa	1.4
1	setosa	1.4
2	setosa	1.3
3	setosa	1.5
4	setosa	1.4
...	...	...
145	virginica	5.2
146	virginica	5.0
147	virginica	5.2
148	virginica	5.4
149	virginica	5.1

150 rows × 2 columns

In [24]:
```
# a subset of the first n rows of the original data
df.iloc[:5,:]
```

Out[24]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

In [25]:
```
#  a subset of the first n columns of the original data
df.iloc[:,:3]
```

Out[25]:

	sepal_length	sepal_width	petal_length
0	5.1	3.5	1.4
1	4.9	3.0	1.4
2	4.7	3.2	1.3
3	4.6	3.1	1.5
4	5.0	3.6	1.4
...	...	...	...
145	6.7	3.0	5.2
146	6.3	2.5	5.0
147	6.5	3.0	5.2
148	6.2	3.4	5.4
149	5.9	3.0	5.1

150 rows × 3 columns

In [26]:
```
# a subset of the first m rows and the first n columns
df.iloc[:4,:5]
```

Out[26]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa

In [27]:
```
# checking the datatype of specific column
df.petal_length.dtype
```

Out[27]:
```
dtype('float64')
```

In [28]:
```
# changing the data type of particular column
df.petal_length.astype(int)
```

Out[28]:
```
0      1
1      1
2      1
3      1
4      1
      ..
145    5
146    5
147    5
148    5
149    5
Name: petal_length, Length: 150, dtype: int64
```

In [29]:
```
# Understanding normalization
df.head()
```

Out[29]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

In [30]:
```
from sklearn import preprocessing
min_max_scaler = preprocessing.MinMaxScaler()
```

In [31]:
```
x=df.iloc[:,:4]
x_scaled = min_max_scaler.fit_transform(x)
df_normalized = pd.DataFrame(x_scaled)
df_normalized
```

Out[31]:

	0	1	2	3
0	0.222222	0.625000	0.067797	0.041667
1	0.166667	0.416667	0.067797	0.041667
2	0.111111	0.500000	0.050847	0.041667
3	0.083333	0.458333	0.084746	0.041667
4	0.194444	0.666667	0.067797	0.041667
...	...	...	...	...
145	0.666667	0.416667	0.711864	0.916667
146	0.555556	0.208333	0.677966	0.750000
147	0.611111	0.416667	0.711864	0.791667
148	0.527778	0.583333	0.745763	0.916667
149	0.444444	0.416667	0.694915	0.708333

150 rows × 4 columns

In [32]:

```
#handling categorical variables
# using get_dummis method of one hot encoding
pd.get_dummies(df, prefix=None, prefix_sep='_',
dummy_na=False,
columns=None,
sparse=False,
drop_first=True, dtype=None)
```

Out[32]:

	sepal_length	sepal_width	petal_length	petal_width	species_versicolor	species_vir
0	5.1	3.5	1.4	0.2	False	
1	4.9	3.0	1.4	0.2	False	
2	4.7	3.2	1.3	0.2	False	
3	4.6	3.1	1.5	0.2	False	
4	5.0	3.6	1.4	0.2	False	
...	...	...	...	...	...	
145	6.7	3.0	5.2	2.3	False	
146	6.3	2.5	5.0	1.9	False	
147	6.5	3.0	5.2	2.0	False	
148	6.2	3.4	5.4	2.3	False	
149	5.9	3.0	5.1	1.8	False	

150 rows × 6 columns

In [33]:
```
# using label encoder
label_encoder = preprocessing.LabelEncoder()
df['Species_n']= label_encoder.fit_transform(df['species'])
df
```

Out[33]:

	sepal_length	sepal_width	petal_length	petal_width	species	Species_n
0	5.1	3.5	1.4	0.2	setosa	0
1	4.9	3.0	1.4	0.2	setosa	0
2	4.7	3.2	1.3	0.2	setosa	0
3	4.6	3.1	1.5	0.2	setosa	0
4	5.0	3.6	1.4	0.2	setosa	0
...	...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	virginica	2
146	6.3	2.5	5.0	1.9	virginica	2
147	6.5	3.0	5.2	2.0	virginica	2
148	6.2	3.4	5.4	2.3	virginica	2
149	5.9	3.0	5.1	1.8	virginica	2

150 rows × 6 columns

In [34]:
```
pd.get_dummies(df, prefix="species", prefix_sep='_',
dummy_na=False,
columns=["species"],
drop_first=True, dtype=None)
```

Out[34]:

	sepal_length	sepal_width	petal_length	petal_width	Species_n	species_versicolor
0	5.1	3.5	1.4	0.2	0	False
1	4.9	3.0	1.4	0.2	0	False
2	4.7	3.2	1.3	0.2	0	False
3	4.6	3.1	1.5	0.2	0	False
4	5.0	3.6	1.4	0.2	0	False
...	...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	2	False
146	6.3	2.5	5.0	1.9	2	False
147	6.5	3.0	5.2	2.0	2	False
148	6.2	3.4	5.4	2.3	2	False
149	5.9	3.0	5.1	1.8	2	False

150 rows × 7 columns

In [ ]: